SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA02 **COURSE NAME: C Programming**

COURSE OUTCOME COVERED

CO1: *Construct structured C programs using constants, variables, data types, and preprocessor directives to address basic computational tasks.* *(BL2, BL3)*

Assessment Tool Weightage: 40%

QUESTIONS: **Total Marks:100**

Annexure A

PSEUDO CODE WRITING

Code of Conduct:

I B.V. Tharun (192525155) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Tharun B.V

Signature of the student:

Annexure A

PSEUDO CODE WRITING

Q1. Given pseudocode:

```
#define DEBUG 0

IF DEBUG
  PRINT intermediate_value
END IF
```

- Identify error in directive usage
- Correct the logic

Q2. #define PI _____

```
DECLARE radius AS float
DECLARE area AS float

INPUT radius

area = _____

PRINT area
```

Fill:

- Constant
- Formula

Q3. #define RATE 1.5

```
DECLARE x AS int = 10
DECLARE result AS float

result = x * RATE

PRINT result
```

Questions:

- Output?
- Why is the result float?

Q4. #define MAX 1000

```
DECLARE temp AS int
INPUT temp

IF temp > MAX
  PRINT "Valid"
ELSE
  PRINT "Invalid"
```

- Identify logical error
- Suggest correct condition

Q5. Given unordered pseudocode steps:

- DEFINE conversion rate
- PRINT result
- INPUT amount
- CALCULATE converted value

Arrange in correct order

Q6.
```
#define MIN 10
#define MAX 50

    DECLARE value AS int
    INPUT value

    _____ (value >= MIN AND value <= MAX)
    PRINT "Within range"

    _____
    PRINT "Out of range"
```

Fill missing control structures

Q7. A billing system includes optional features such as discount calculation and tax computation. Depending on the business requirement, these features may be enabled or disabled at compile time. The program should calculate the total bill amount based on input price and quantity. If discount and tax features are enabled, they should be applied using predefined constants; otherwise, they should be skipped. The system must ensure that floating-point precision is maintained during calculations.

Write pseudocode using preprocessor directives to enable/disable features, define constants for discount and tax rates, and use appropriate data types.

Q8. #define TAX 5%

DECLARE price AS int
DECLARE total AS int

total = price + (price * TAX)

PRINT total

- Identify errors
- Correct data types and constant definition

9. A temperature conversion program supports multiple modes such as Celsius to Fahrenheit and Fahrenheit to Celsius. The mode of operation should be selected at compile time using preprocessor directives. The program should use appropriate floating-point data types to ensure accuracy and define conversion formulas using symbolic constants wherever applicable. The final output should be displayed with two decimal precision.
Write pseudocode using preprocessor directives for mode selection, constants for conversion formulas, and proper data types.

10. #define SIZE 5

DECLARE arr[SIZE]
INPUT elements

- Extend pseudocode to:
  - Find sum
  - Compute average

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

2. **Pseado Code writing :-**

1. **Error :-** It debug is incorrect because DEBUG is a preprocessor constant

Correct Pseado

```
# DEBUG
# if DEBUG
  Print intermediate-value
# endit
```

2. **Constant :-** #define PI=3.14

Formula :- area = PI * radius * radius

3. **output** = 15.0

Reason = RATE is a floating-point constant (1.5) so the multiplication produces a float.

4. **Logical error :-** The condition is reversed

```
:- if temp<max
   Print "valid"
   else
      Print "Invalid"
   ENDIt.
```

5. **Correct order :-**

1. DEFINE conversion rate
2. Input amount
3. Calculate converted value

```
#define MIN 10
#define MAX 50
Declare value as int
input value
if (value >= Min and value <= Max)
else print "out of range"
endit
```

```
#define DIS 10
#define TAX 5
input price, qty
total = price, ity

print total.
```

Errors :- #define TAX 5% is invalid, price & total

```
#defin TAX 0.05
Declare price as float
Declare total float
print total
```

```
#define mode
input temp
# if mode
print (temp * 9/5) + 12
# else
print (temp - 32) * 5/9
```